

# MB Note

## 1 Population Models for Single Species

**Logistic Growth Model (Population): Model:**  $\frac{dN}{dt} = rN(1 - \frac{N}{K})$  N: population at time  $t$ ; K: carrying capacity (depends on resources);  $r, K > 0$ .

**Sol:**  $N(t) = \frac{N_0 K e^{rt}}{K + N_0(e^{rt} - 1)} \rightarrow K$  **Equilibrium points:**  $N = 0$  (unstable),  $N = K$  (stable). **Quickest:**  $N = \frac{K}{2}$  (<: 凹; >: 凸)

**General Population Model: Model:**  $\frac{dN}{dt} = f(N)$   $f(N)$ : nonlinear function. **No limit cycle periodic sol.**

·  $N^*$  equilibrium point ( $f(N^*) = 0$ )  $\Rightarrow$   $^1 f'(N^*) < 0$  (stable for small perturbations);  $^2 f'(N^*) > 0$  (unstable) **Time-Independent:** Equilibrium |  $\frac{dN}{dt} = 0 \Rightarrow N(t) = N(t - T) = N^*$ .

**Linearise at  $N^*$ :** Consider  $N(t) = N^* + n(t)$ ,  $|n(t)| \ll 1 \Rightarrow \frac{dn}{dt} = \frac{dN}{dt} = (\text{单})f(N^*) + f'(N^*)n(t) = f'(N^*)n(t)$  (Delay) 带入  $f \approx$  (求 2 次以上) **Small Perturbations:**  $n(t)$

**Stability  $N^*$ :** 计算扰动  $n(t)$  的  $\frac{dn}{dt} \rightarrow$  假设  $n(t) = e^{\lambda t}$  |  $ce^{\lambda t}$  求解  $\lambda$  隐式表达  $\rightarrow \lambda = u + iw$  带入解出  $u$  和  $w$ .  $\rightarrow u < 0$  asy.stable;  $u > 0$  unstable. **Find  $T_c$ :** 令 (delay)  $T = 0$ , 验稳定性, 如稳,  $u = 0 (\lambda = iw)$  为  $T_c$

**Delay Model: Model:**  $\frac{dN}{dt} = f(N(t), N(t - T))$  T: delay time  $> 0$  周期解 | 与初始值无关 ps: To consider e.g. gestation period, maturation time etc.

$^1$  Ext. of Logistic Model:  $\frac{dN}{dt} = rN(t) \left[ 1 - \frac{N(t-T)}{K} \right]$  (用 "ave. pop." 估计, 准用 "Convolution Type")  $\Rightarrow$   $^2$  Dimensionless:  $\frac{dN^*}{dt^*} = N^*(t^*) [1 - N^*(t^* - T^*)]$  ( $N^* = \frac{N}{K}$ ,  $t^* = rt$ ,  $T^* = rT$ )

$^{1,2}$  Periodic sol: stable limit cycle  $N(t + t_p) = N(t)$ ,  $t_p$ : period. **Amplitude:**  $rT$  **Equilibrium Points:**  $N = 0$  |  $N^* = 0$  (unstable);  $N = K$  |  $N^* = 1$  (Stable:  $0 < T^* = rT < \frac{\pi}{2}$ ; Unstable:  $T^* > \frac{\pi}{2}$ )

**Continuous Population Model (EN): Model:**  $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - EN = f(N)$   $r, K, E > 0$ ; EN: harvesting rate/unit time; E: effort measure.

**Equilibrium:**  $N = 0$  (unstable);  $N = N_h(E) = K(1 - \frac{E}{r}) > 0$  (need  $E < r$ , Stable) **Yield:**  $Y(E) = EN$  ·  $E > r$ : die out; else: **Max Yield:**  $Y_M = \frac{rK}{4}$  ( $E = \frac{r}{2}$ ) **Steady State:**  $N_h|_{Y_M} = \frac{K}{2}$

**Continuous Population Model ( $Y_0$ ): Model:**  $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - Y_0 = f(N; r, K, Y_0)$   $Y_0$ : constant yield rate/unit time.

**Max Yield:**  $Y_M = \frac{rK}{4}$  **Equilibrium:**  $^1 0 < Y_0 < Y_M$ : Two positive points (unstable(小) and stable(大)). **Sensitivity:** 对于 fluctuation 此模型和 EN 模型都不稳定, 但此模型更敏感 sensitive.

## 2 Discrete Population Models

**Discrete Model: Model:**  $N_{t+1} = N_t F(N_t) = f(N_t)$  · e.g. If  $F(N_t) = r > 0$ : sol will decays|grows geometrically for  $r < 1$  |  $r > 1$ .

**Max|Min:** It's expected has max at  $N_m$  ( $N_{\max} = f(N_m)$ ;  $\frac{df}{dN_t}|_{N_m} = 0$ ); then  $N_{\min} = f(N_{\max}) = f^2(N_m)$ ; If  $N_{\min} \leq 1$ , **species extinction.** **Equilibrium:**  $N^* = f(N^*)$

**Eigenvalue:**  $\lambda = f'(N^*)$   $N^* > N_m$ :  $|\lambda| < 1$ : stable (critical bifurcation);  $|\lambda| = 1$ : periodic (attracting (+: tangent, -: period-doubling) bifurcation);  $|\lambda| > 1$ : unstable. (chaotic)

**Perturbations at  $N^*$ :** A.(stable) 扰动  $\rightarrow 0$  by a1.  $0 < \lambda < 1$  ↓ monotone; a2.  $-1 < \lambda < 0$  decreasing oscillations; B.(unstable) 扰动  $\rightarrow \infty$  by b1.  $\lambda > 1$  ↑ monotone; b2.  $\lambda < -1$  increasing oscillations.

**Periodic:**  $^1$  Equilibrium:  $f^p(N^*) = N^*$  (指迭代  $p$  次) is  $p$ -periodic for original eq.  $^2 \forall$  even  $p$ -periodic  $N^*$  with  $\lambda = -1$  branches into  $2p$ -periodic  $N^*$ ; 后续分支间隔越来越短.

**Sarkovskii's Theorem:** If a 3-periodic  $N^*$  exists, then  $\forall p \geq 1$ ,  $p$ -periodic  $N^*$  exists. (i.e. chaos) **Verhulst:**  $N_{t+1} = rN_t(1 - \frac{N_t}{K})$ ;  $r, K > 0$ . Better: **Ricker curve:**  $N_{t+1} = N_t e^{r(1 - \frac{N_t}{K})}$

**Linearise at  $N^*$ :** Consider  $N_t = N^* + n_t$ ,  $|n_t| \ll 1 \rightarrow N^* + n_{t+1} = \dots = f(N^* + n_t) \approx f(N^*) + f'(N^*)n_t \rightarrow n_{t+1} = f'(N^*)n_t = \lambda n_t \rightarrow n_t = \lambda^t n_0$ . 分析  $t \rightarrow \infty$

## 3 Bifurcation

**Bifurcation:** Qualitative changes in dynamics are called *bifurcations*. (当平衡点发生变化时) **Bifurcation Diagram:** y 轴: 平衡点; x 轴: 控制参数; 虚线 unstable, 实线 stable. 画图判别

**Tangency Condition:** If  $\dot{x} = f(x, r)$ , at bifurcation point  $(x^*, r_c)$ :  $\frac{\partial f}{\partial x}|_{(x^*, r_c)} = 0$  (Basic:  $f(x^*, r_c) = 0$ ) · At bifurcation point: 2 equilibrium **coalesce.**

**Normal Forms|Saddle-Note:**  $\dot{x} = r \pm x^2$  ( $r_c = 0$ ;  $r \neq 0$  时两个  $x^*$ ) **Normal Forms|Transcritical:**  $\dot{x} = rx - x^2$  ( $r_c = 0$ ; if  $r \neq 0$ ,  $x^* = \{0, r\}$ , 大 stable, 小 unstable)

**Normal Forms|Pitchfork:**  $\dot{x} = rx - x^3$  ( $r_c = 0$ ; if  $r < 0$ ,  $x^* = 0$  stable; if  $r = 0$ ,  $x^* = 0$  stable (weakly)|critical slowing down); if  $r > 0$ ,  $x^* = \pm\sqrt{r}$  stable,  $x^* = 0$  unstable) · Has symmetry.

**Normal Forms|Subcritical Pitchfork:**  $\dot{x} = rx + x^3$  ( $r_c = 0$ ; if  $r < 0$ ,  $x^* = 0$  stable,  $\pm\sqrt{-r}$  unstable; if  $r = 0$ ,  $x^* = 0$  stable (weakly); if  $r > 0$ ,  $x^* = 0$  unstable) (其他: linearisation 后对比 normal forms 判别)

**Imperfect Bifurcation:**  $\dot{x} = rx - x^3 + h$  (codimension-2 bifurcation) If  $r \leq 0$ ,  $x^*$  unique. If  $r > 0$ ,  $|h| > h_c$ ,  $x^*$  unique;  $|h| < h_c$ ,  $x^*$  three. (两边 stable, 中 unstable). Bifurcation points:  $h = \pm h_c = \pm \frac{2r}{3\sqrt{3}}$ . **cusp point:**  $(r, h) = (0, 0)$

## 4 Appendix

$y' + p(x)y = q(x)$ : **Integrating Factor:**  $I(x) = e^{\int p(x)dx} \Rightarrow I(x)y = \int I(x)q(x)dx$  **Taylor Expansion:**  $f(x, r)$  at  $(x_0, r_0)$ :  $f(x, r) = f(x_0, r_0) + (x - x_0)f_x|_{(x_0, r_0)} + (r - r_0)f_r|_{(x_0, r_0)} + \frac{1}{2} \dots$

**Binomial Series:**  $(1 + x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \dots$  **Linerisation:**  $f(x + \Delta x) = f(x) + f'(x)\Delta x + \dots$   $e^{bx} = 1 + bx + \frac{(bx)^2}{2!} + \dots$